
CODE OF BOOTH MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/booth/booth_mul.vhd [VHDL]

```
1  -----
2  --
3  -- Company:
4  -- Engineer:
5  --
6  -- Create Date: 09/10/2026 04:26:32 PM
7  -- Design Name:
8  -- Module Name: booth_mul - Behavioral
9  -- Project Name:
10 -- Target Devices:
11 -- Tool Versions:
12 -- Description:
13 --
14 -- Dependencies:
15 --
16 -- Revision:
17 -- Revision 0.01 - File Created
18 -- Additional Comments:
19 --
20 -----
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 use IEEE.NUMERIC_STD.ALL;
28 --library UNISIM;
29 --use UNISIM.VComponents.all;
30
31 entity booth_mul is
32     Port ( a : in signed (3 downto 0);
33           b : in signed (3 downto 0);
34           p : out signed (7 downto 0));
35 end booth_mul;
36
37 architecture Behavioral of booth_mul is
38 begin
39
40     process (a , b)
41         variable q_pair : std_logic_vector(1 downto 0);
42         variable acc : signed (3 downto 0);
43         variable q : signed (3 downto 0);
44         variable total : std_logic_vector (8 downto 0);
45     end process;
```

46	variable w : integer := 3;
47	
48	begin
49	q_pair := "00";
50	acc := "0000";
51	q := b;
52	total := std_logic_vector(acc) & std_logic_vector(q) & q_pair(0);
53	
54	for i in 0 to w loop
55	
56	q_pair := total(1 downto 0);
57	
58	case(q_pair) is
59	when "01" => acc := acc + a;
60	when "10" => acc := acc -a ;
61	when others => null ;
62	end case;
63	
64	total := std_logic_vector(acc) & std_logic_vector(q) & q_pair(0);
65	total := total(8) & total(8 downto 1);
66	acc := signed(total(8 downto 5));
67	q := signed(total(4 downto 1));
68	
69	end loop;
70	
71	p <= signed(total(8 downto 1));
72	
73	end process;
74	
75	end Behavioral;

CODE OF TEST BENCH FOR BOOTH MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/booth/booth_mul_tb.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 09/10/2026 04:26:32 PM
6	-- Design Name:
7	-- Module Name: booth_mul_tb - Behavioral
8	-- Project Name:
9	-- Target Devices:
10	-- Tool Versions:
11	-- Description:
12	--
13	-- Dependencies:
14	--
15	-- Revision:

```

16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 --
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity booth_mul_tb is
35 -- Port ( );
36 end booth_mul_tb;
37
38 architecture testing of booth_mul_tb is
39
40
41     component booth_mul is
42         Port ( a : in signed (3 downto 0);
43               b : in signed (3 downto 0);
44               p : out signed (7 downto 0));
45     end component;
46
47
48     signal a : signed(3 downto 0);
49     signal b : signed(3 downto 0);
50     signal p : signed(7 downto 0);
51
52 begin
53
54
55     booth_multiplier_test : booth_mul port map(
56         a => a,
57         b => b,
58         p => p
59     );
60
61
62     process
63
64     begin

```

65	a <= "0101";
66	b <= "1010";
67	
68	wait for 10ns;
69	
70	a <= "0011";
71	b <= "1100";
72	
73	wait for 10ns;
74	
75	a <= "1010";
76	b <= "1100";
77	
78	wait for 10ns;
79	
80	a <= "0001";
81	b <= "1000";
82	
83	wait for 10ns;
84	
85	a <= "1110";
86	b <= "1011";
87	
88	wait for 10ns;
89	
90	a <= "1111";
91	b <= "1111";
92	
93	wait for 10ns;
94	
95	a <= "0000";
96	b <= "0000";
97	
98	wait for 10ns;
99	
100	a <= "1010";
101	b <= "1111";
102	
103	wait for 10ns;
104	
105	a <= "0001";
106	b <= "1111";
107	
108	wait for 10ns;
109	
110	a <= "1010";
111	b <= "1111";
112	
113	
114	wait;

115	
116	end process;
117	
118	
119	
120	end testing;

SCREENSHOT OF CODE

The screenshot shows the Vivado IDE interface with the 'PROJECT MANAGER' tab selected. The 'booth_mul.vhd' file is open in the editor. The code defines an entity 'booth_mul' with two 3-bit signed inputs 'a' and 'b', and a 7-bit signed output 'p'. The architecture is behavioral and implements a Booth multiplier. It uses a process block to calculate the product of 'a' and 'b' using a series of shifts and additions. The code is as follows:

```
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 use IEEE.NUMERIC_STD.ALL;
28 --library UNISIM;
29 --use UNISIM.VComponents.all;
30
31 entity booth_mul is
32   Port ( a : in signed (3 downto 0);
33         b : in signed (3 downto 0);
34         p : out signed (7 downto 0));
35 end booth_mul;
36
37 architecture Behavioral of booth_mul is
38   begin
39     process (a, b)
40       variable q_pair : std_logic_vector(1 downto 0);
41       variable acc : signed (3 downto 0);
42       variable q : signed (3 downto 0);
43       variable total : std_logic_vector (8 downto 0);
44       variable w : integer := 3;
45
46     begin
47       q_pair := "00";
48       acc := "0000";
49       q := b;
50       total := std_logic_vector(acc) & std_logic_vector(q) & q_pair(0);
51       for i in 0 to w loop
52         q_pair := total(1 downto 0);
53         case(q_pair) is
54           when "01" => acc := acc + a;
55           when "10" => acc := acc - a;
56           when others => null;
57         end case;
58         total := std_logic_vector(acc) & std_logic_vector(q) & q_pair(0);
59         total := total(8) & total(8 downto 1);
60         acc := signed(total(8 downto 5));
61         q := signed(total(4 downto 1));
62       end loop;
63       p <= signed(total(8 downto 1));
64     end process;
65 end Behavioral;
```

Figure 1: Design

The screenshot shows the Vivado IDE interface with the 'PROJECT MANAGER' tab selected. The 'booth_mul_tb.vhd' file is open in the editor. The code defines a testbench for the 'booth_mul' entity. It instantiates the 'booth_mul' component and applies test vectors to the inputs 'a' and 'b'. The testbench is as follows:

```
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity booth_mul_tb is
35   Port ( );
36 end booth_mul_tb;
37
38 architecture testing of booth_mul_tb is
39   component booth_mul is
40     Port ( a : in signed (3 downto 0);
41           b : in signed (3 downto 0);
42           p : out signed (7 downto 0));
43   end component;
44
45   signal a : signed(3 downto 0);
46   signal b : signed(3 downto 0);
47   signal p : signed(7 downto 0);
48
49   begin
50     booth_multiplier_test : booth_mul port map(
51       a => a,
52       b => b,
53       p => p
54     );
55
56     process
57       begin
58         a <= "0101";
59         b <= "1010";
60         wait for 10ns;
61         a <= "0011";
62         b <= "1100";
63         wait for 10ns;
64         a <= "1010";
65         b <= "1100";
66         wait for 10ns;
67         a <= "0001";
68         b <= "1000";
69       end process;
70 end testing;
```

Figure 2: Tb

WAVEFORM

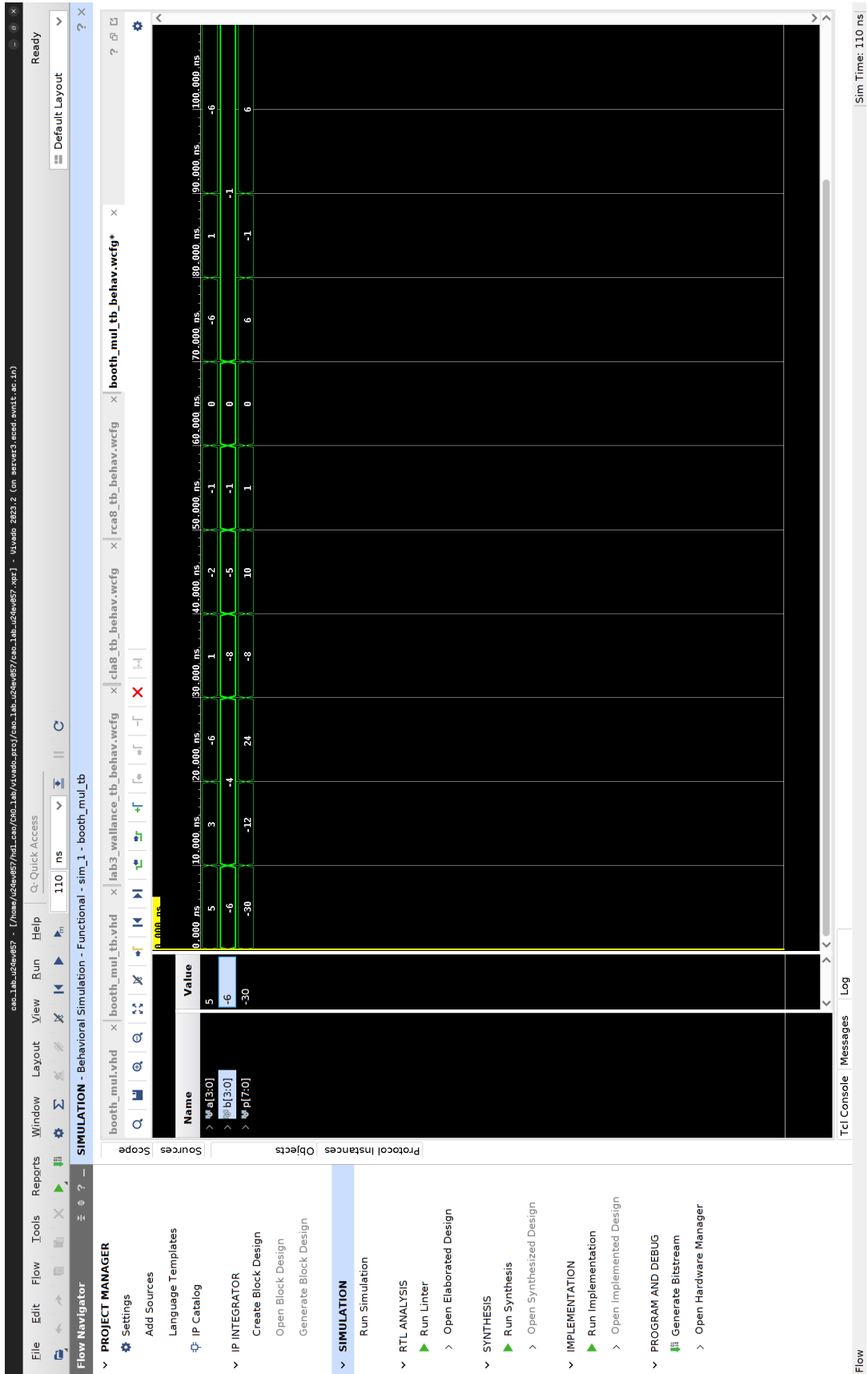


Figure 3: Waveform